

MARYBOROUGH, Australia

WMO: 945670

Lat: 25.5161S

Lon: 152.7156E

Elev: 11

StdP: 101.20

Time Zone: 10.00 (E10)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	5.2	6.9	-1.2	3.4	17.2	0.7	4.0	14.9	9.2	18.9	8.4	19.0	1.0	230	0.350

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	8.5	32.7	23.8	31.6	23.3	30.6	22.9	25.7	29.8	25.1	28.9	24.5	28.2	4.5	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
24.7	19.7	27.6	24.0	19.0	26.9	23.4	18.2	26.5	79.5	30.0	76.8	28.9	74.4	28.2	30.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.3	8.4	7.5	1.8	35.4	1.0	1.2	1.0	36.2	0.4	37.0	-0.2	37.6	-0.9	38.5		
			1.2	27.4	1.1	1.3	0.4	28.3	-0.2	29.1	-0.8	29.8	-1.6	30.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	21.0	25.6	25.4	24.3	21.8	18.8	16.4	15.6	16.3	19.1	21.2	23.3	24.8
	DBStd	4.07	1.71	1.66	1.50	1.66	2.18	2.33	2.15	2.20	2.12	2.02	1.99	1.86
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	259	0	0	0	1	21	64	88	69	14	1	0	0
	CDD10.0	4022	483	430	444	354	272	193	173	196	273	348	398	458
	CDD18.3	1240	224	197	186	106	34	7	3	7	37	91	148	200
	CDH23.3	8698	1844	1517	1189	481	115	17	18	57	263	553	1036	1609
	CDH26.7	2425	618	489	311	64	5	0	1	2	29	96	282	528
Wind	WSAvg	3.1	3.7	3.7	3.5	3.2	2.7	2.6	2.5	2.7	3.0	3.3	3.4	3.5
Precipitation	PrecAvg	1070	149	171	123	76	81	57	49	40	33	76	96	125
	PrecMax	1668	504	900	519	291	344	348	406	108	131	215	302	325
	PrecMin	617	6	12	14	18	2	0	0	0	1	8	9	17
	PrecStd	273	104	149	95	61	59	64	63	26	32	43	61	70
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	34.0	33.8	32.5	29.7	27.3	25.2	25.0	26.5	29.1	30.9	32.2	34.0
		MCWB	24.8	24.2	23.7	21.5	19.6	18.7	16.8	17.6	18.2	20.2	21.9	23.4
	2%	DB	32.4	32.2	30.8	28.4	25.7	23.7	23.5	24.9	27.4	28.8	30.7	32.1
		MCWB	24.2	24.3	23.2	21.1	18.8	17.7	16.3	16.9	18.4	20.3	21.9	23.3
	5%	DB	31.1	30.8	29.6	27.2	24.6	22.6	22.2	23.6	26.1	27.6	29.6	30.8
		MCWB	23.8	23.7	22.6	20.5	18.3	17.1	15.7	16.1	17.9	19.9	21.5	22.8
	10%	DB	29.9	29.6	28.4	26.2	23.5	21.5	21.0	22.3	24.9	26.4	28.3	29.6
		MCWB	23.0	23.1	22.2	20.2	17.9	16.5	15.3	15.5	17.7	19.3	20.9	22.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	27.0	26.7	25.6	24.0	22.2	20.2	19.8	20.2	21.7	23.2	24.7	26.2
		MCDB	31.2	31.3	28.7	27.5	24.5	22.9	22.6	24.1	25.5	28.1	29.6	30.6
	2%	WB	25.7	25.6	24.7	22.6	21.0	19.2	18.4	18.9	20.8	22.2	23.8	25.1
		MCDB	29.7	29.8	27.9	25.9	23.1	21.7	21.2	22.2	24.8	26.1	28.0	29.2
	5%	WB	25.1	24.9	24.0	21.9	20.0	18.4	17.3	18.0	19.9	21.4	23.0	24.3
		MCDB	28.7	28.8	27.4	25.0	22.9	21.1	20.2	21.3	23.5	25.1	26.8	28.1
	10%	WB	24.4	24.2	23.3	21.1	19.1	17.5	16.4	17.1	19.1	20.6	22.2	23.6
		MCDB	27.9	27.7	26.7	24.5	22.1	20.3	19.4	20.3	22.6	24.3	26.0	27.3
Mean Daily Temperature Range	5% DB	MDBR	8.5	8.2	8.4	9.1	10.1	10.6	11.9	12.9	12.2	10.4	9.8	9.2
		MCDBR	10.3	10.1	10.3	10.6	11.3	11.2	13.3	14.4	13.9	12.1	11.4	10.9
		MCWBR	3.4	3.4	3.7	4.3	5.2	5.8	6.7	7.0	6.0	4.7	4.1	3.7
	5% WB	MCDBR	8.3	8.4	8.3	8.6	8.4	9.0	10.0	11.1	10.4	9.6	9.1	8.7
		MCWBR	3.4	3.4	3.5	4.0	4.5	5.0	5.9	6.3	5.1	4.2	4.0	3.7
Clear-Sky Solar Irradiance	taub	0.391	0.385	0.368	0.343	0.323	0.317	0.307	0.318	0.356	0.380	0.378	0.394	
	taud	2.508	2.529	2.573	2.626	2.633	2.640	2.651	2.605	2.489	2.455	2.496	2.480	
	Ebn at Noon	953	943	931	912	890	874	899	924	928	937	958	952	
	Edh at Noon	115	110	100	87	79	75	76	87	107	117	115	118	
All-Sky Solar Radiation	RadAvg	6.56	5.95	5.20	4.59	3.86	3.29	3.76	4.64	5.65	6.18	6.67	6.69	
	RadStd	0.50	0.51	0.41	0.34	0.19	0.25	0.32	0.35	0.42	0.52	0.64	0.72	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	-0.43	-0.56	N/A	N/A	N/A	N/A
(47)	Regional (16 neighbors)	+0.30	N/A	N/A	N/A	N/A	N/A	N/A	-80	+208	+131

Nomenclature: See separate page